

Object Cluster Registration of Dissimilar Rooms Using Geometric Spatial Affordance Graph to Generate Shared Virtual Spaces

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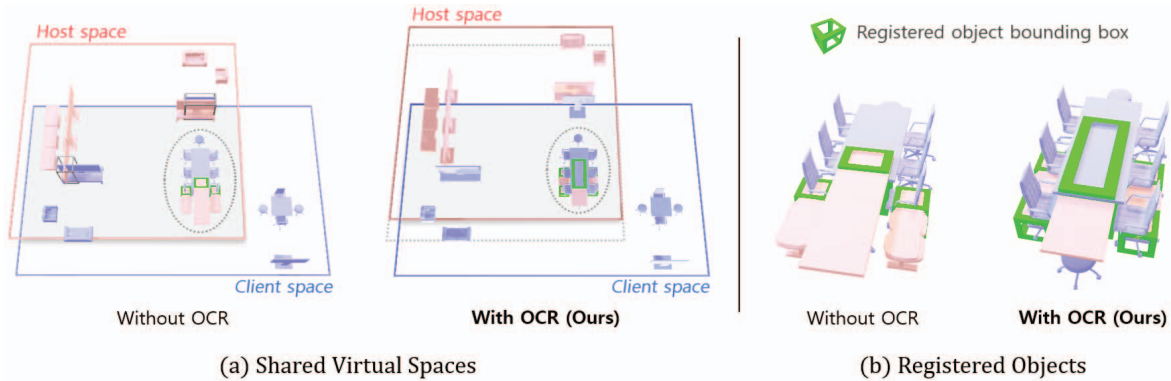


Figure 1: This study proposes a shared virtual space considering optimal object clusters through Object Cluster Registration (OCR) using Geometric Spatial Affordance Graph (GSAG). (a) is the shared virtual space created when applying the existing method that aims to maximize the area without OCR and the proposed OCR using GSAG. (b) is registered objects for multi-object interaction in each method. When OCR is used, more and wider co-interactable objects (highlighted green box) are registered as the same cluster.

ABSTRACT

We propose Object Cluster Registration (OCR) using Geometric Spatial Affordance Graph (GSAG) to support user interaction with multiple objects in a shared space generated from two dissimilar rooms. Previous research on generating a shared virtual space from dissimilar real spaces has only reflected the information of individual objects and aimed at maximizing the area of the shared space. This led to limited interactions relying on the singular affordances of objects, neglecting to consider the usability and effectiveness of the generated shared spaces. The proposed OCR with GSAG, which considers the relationship between objects based on facing formation, extracts optimal object cluster pairs to align dissimilar rooms in generating shared virtual spaces. In an evaluation study involving 100 multi-cluster space pairs, applying OCR using GSAG showed greater effectiveness in preserving object correlations compared to cases where OCR was not used. Furthermore, the size of the shared space did not significantly differ between the two methods. This suggests that factoring in the relationship between objects does not compromise the objective of maximizing the shared virtual space. The proposed method is expected to serve as a foundation for generating shared virtual spaces that are more user-oriented and efficient by facilitating a wider range of collaborative activities for remote users in dissimilar real spaces with varied configurations.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction

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(HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

With various video conferencing platforms such as Zoom¹, Microsoft Teams², and Google Meet³, users can collaborate while engaging in face-to-face interaction despite being physically distant from one another. However, there is an increasing demand for communication beyond the confines of a 2D screen that closely resembles meeting and working together in person. For this, advancements in Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR) technology have enabled remote collaboration in shared virtual spaces based on the real environment, such as meeting rooms, workspaces, and studios, which involves a wide range of 3D interactions [30, 32, 51]. Such collaboration scenarios call for the need to consider, incorporate, and align the physical environment of each user located remotely in different spaces to support these interactions. This has led to research focusing on the spatial aspects of MR remote collaboration for more realistic and immersive experiences. [23, 37, 49].

Efforts to replicate real-world interactions in shared virtual spaces have explored resolving spatial discrepancies by adjusting avatar positions [51] or aligning common spatial features [13], often overlooking the unique configurations of individual spaces. Previous studies focusing on this aspect emphasized one-to-one semantic object similarity across user spaces, neglecting intra-space object relationships [17]. Moreover, quantitative measures for object relationships, critical for addressing spatial disparities in shared virtual space generation, have been underexplored. In addition, methods to generate shared virtual spaces from multiple dissimilar spaces have primarily focused on maximizing the area of the outcome. This has limited the range of interactions that the shared virtual space can afford to users collaborating in it.

¹<https://zoom.us/>

²<https://www.microsoft.com/microsoft-teams/group-chat-software/>

³<https://meet.google.com/>

To address this gap in research, we propose a new data structure called Geometric Spatial Affordance Graph (GSAG) based on the facing formation between objects as shown in Fig. 1, reflecting both their geometric and semantic attributes [41]. Using GSAG, we devise an Object Cluster Registration (OCR) method, which identifies and registers interactable object clusters based on their affordances. By connecting object clusters through OCR, the generated shared virtual space allows for a wider range of interactions. The technical advantage of GSAG is that it does not view objects as individual entities by considering the facing direction and distance of the furniture and determining relationships through these. The technical advantage of OCR lies in the ability to ensure a wider range and greater number of shared objects that can be used together in a shared virtual space, as illustrated in Fig. 1(b).

We demonstrate the capacity of OCR by comparing the Registered Edge-IoU (REI) and four different types of Registered Area-IoU (RAI) of the shared virtual spaces generated by it with existing generated methods that do not apply OCR using GSAG. 100 shared virtual spaces were generated from a combination of 10 host spaces and 10 client spaces, which were each created by the random arrangement of five single-cluster spaces based on real-space data. The results show that OCR using GSAG can effectively form a shared virtual space in which object relationships are preserved. Additionally, when compared to an existing algorithm that aims to maximize the area of the shared space, there was no significant decrease in the total area: Rather, a significant increase in the furniture area was found. This indicates that considering object relationships in the generation of shared virtual spaces to support a wider variety of interactions does not compromise the goal of maximizing the area and can even be beneficial towards this end. In addition, applicability to real space was confirmed by employing the proposed GSAG method to the Matterport3D dataset [5]. The contributions of this study are as follows: First, we propose GSAG, a novel approach that represents object relationships using facing formations based on geometric spatial affordances. Second, we propose OCR using GSAG as a new approach to finding optimal object cluster pairs that minimize geometric dissimilarity while maximizing the weighted sum of edges in the matched clusters. Finally, we suggest Registered Edge-IoU (REI) as a novel quantitative measure of multi-object relationships. Our study is the first attempt to view space as a set of relationships between objects based on geometric spatial affordance, rather than a set of individual objects that can be interacted with, to enable a wide variety of collaborative activities in a shared virtual space. Our method is beneficial for dynamic remote collaboration that leverages the user's own physical environment as a combination of multiple objects identified through their affordances and relationships. The study presents new directions for creating shared spaces between asymmetric spaces that enable higher levels of immersion and social presence among users in various MR remote collaboration scenarios.

2 RELATED WORKS

2.1 Defining Inter-Object Relationships for User-Object Interaction

As direct interaction with objects in space leads to a more immersive user experience [14], allowing various interactions in the shared virtual space is essential for remote collaboration. To this end, Keshavarzi et al. [17, 18] identified and matched objects of the same affordance in different user spaces in the generation of a shared virtual space. However, to support a wider variety of user interactions that involve not only single objects but also multiple objects at the same time, the relationship between objects placed within a user space must be considered as well. In this light, studies on space-adaptive interfaces have considered the relationship between the virtual interface and physical objects in various room conditions [6, 11, 34]. However, they have not explored how physical objects are placed and put into function with regard to other objects

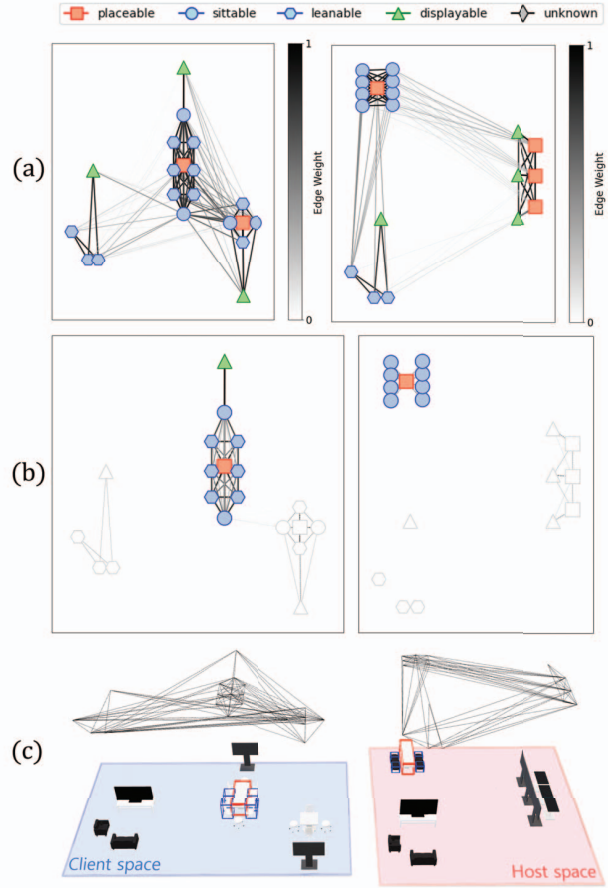


Figure 2: This figure shows the entire process of creating a shared virtual space through OCR using GSAG. The first picture (a) is the result of different spaces expressed in GSAG. The second picture (b) shows the registered optimal object cluster between them. The third picture (c) depicts registered objects in the shared virtual space.

within the same space in including objects in the interaction scheme.

In order to express and update inter-object relationships, scene graph data structures have been devised [1, 15, 27, 39, 42]. They have largely been employed to automatically arrange or move objects in a room based on their relationships for scene synthesis [28, 43–45]. In addition, there have been studies using scene graphs to identify hyper-relationships between furniture or to synthesize scenes through heatmaps that consider spatial graphs and affordances [33, 52]. However, these studies aimed at improving the feasibility of generated scenes and have not addressed the relationships between objects to devise user-object interactions. In this study, we employ the scene graph structure to express relationships between physical objects to support interaction with multiple objects.

2.2 Facilitating Interaction Between Dissimilar Rooms

Overcoming dissimilarities between multiple user spaces is one of the biggest challenges in facilitating object-user and user-user interactions for MR remote collaboration in a shared virtual space. The dissimilarity between spaces of collaborating users is largely caused by differences in the arrangement of objects in each space. Therefore, it is important to understand the logic of these arrangements. Jo et al. [16] devised a one-to-one matching method for multiple chairs

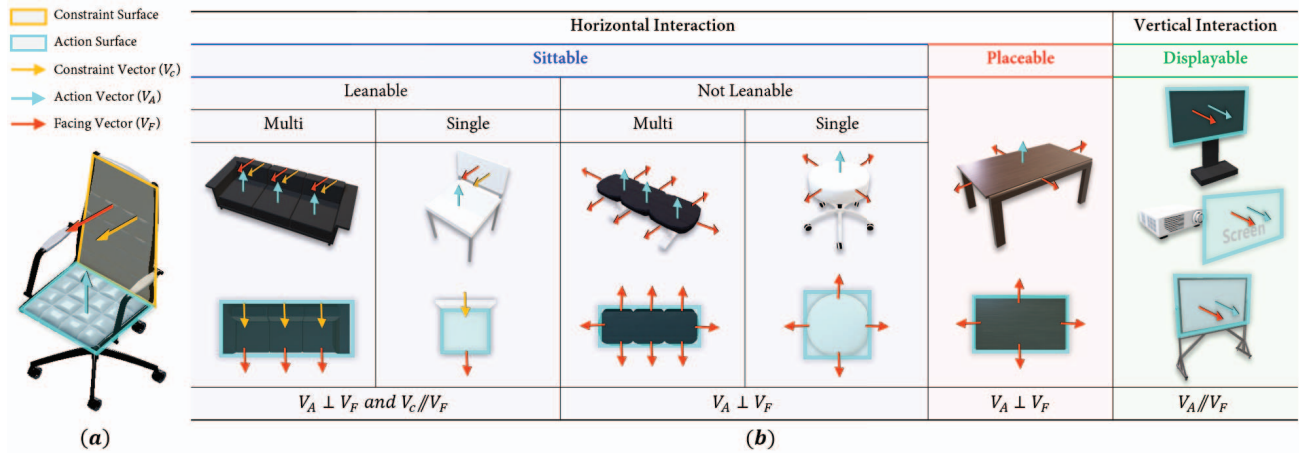


Figure 3: (a) shows the constraint surface, action surface, constraint vector (V_C), action vector (V_A), and facing vector (V_F) of the chair. (b) shows five different types of facing-vector generation conditions through geometric spatial affordances. Depending on the three basic affordances (sittable, placeable, and displayable) and the leanable affordance, geometric transformations are computed differently to determine the facing vector.

of various arrangements. De Bauw et al. [8] mapped two desks with different sizes and shapes located in separate spaces based solely on their semantic property. In addition, Fidalgo et al. [13] suggested remapping whole spaces to match horizontal or vertical planes in dissimilar spaces. While these methods can be applied to align two highly dissimilar spaces, they do not consider the geometric dissimilarity in matching objects, which can decrease the usability of the shared virtual space [10, 36].

On the other hand, many studies that have aligned dissimilar spaces for MR remote collaboration transform the positions or poses of avatars instead of objects or the spaces themselves [7, 31, 51]. Other studies have redirected hands to facilitate collaboration using objects [12, 50]. However, these approaches are limited in that changes to the user's coordinate axis or disjointed actions may escalate the frequency of resets and interrupt users. This limitation extends to methods aligning objects and spaces through modification as well [10, 36]. To solve these problems, we identify common areas for a shared virtual space without modifying the position of avatars, objects, or the spaces themselves. Instead, we cluster objects within a space based on their affordances and determine optimal cluster pairs by considering geometric dissimilarity and REI, which calculates how many relationships are actually secured in a common area within a given room, to reduce the possibility of resets between dissimilar spaces.

2.3 Generating Shared Virtual Spaces

One of the most important goals in creating a virtual space is to increase its area, as virtual spaces can be larger than the real space they are experienced in. In particular, much research has been conducted on Redirected Walking (RDW), a VR locomotion technique that allows users to walk in a virtual space that is larger than the real space by adjusting the user's walking speed [35]. There have been many studies that apply RDW to collaboration between multiple co-located users [3, 9, 24, 26, 29]. The goal of maximizing the area of the shared virtual space has been no exception in these cases. The approach to establishing a shared virtual space centered around creating an overlapping region to ensure a space usable by all parties involved [25, 40]. Xu et al. [48] further employed RDW methods in a remote collaboration context. Other methods aimed at increasing the area of the shared virtual space include rescaled walking [19, 21] and moving physical objects to secure a larger space [17].

As a result, numerous studies have assessed the outcome of various methods for shared virtual space generation by employing the size of its area as a quantitative measure. Conversely, studies on spatial indicators that evaluate or analyze the shared virtual space with regard to user interaction have been relatively scarce. As RDW and rescaled walking techniques are affected not only by size but also spatial configuration, several measures concerning the latter have been proposed [2, 20]. Williams et al. [46, 47] presented Complexity Ratio (CR) and Environment Navigation Incompatibility (ENI) as such metrics. However, objects were perceived as mere obstacles in these measures because they were proposed in RDW studies focused on spatial navigation. While Shin et al. [38] employed spatial complexity as a factor of affordance that impacts narrative experiences in AR, it was not presented as a quantitative measure. In addition, Kim et al. [22] used the matching rate between two objects' edges as an indicator to evaluate potential interactions with objects in remote collaboration. Although they proposed a quantitative means to assess the relationship between two objects in different user spaces, systematic measures to understand the relationship between multiple objects within a space have yet to be devised.

3 METHODOLOGY

This study proposes an Object Cluster Registration (OCR) to find co-interactable object cluster pairs with the Geometric Spatial Affordance Graph (GSAG) between asymmetric spaces for shared virtual space generation (Fig. 2). The process consists of two steps: GSAG generation of a single room and OCR between asymmetric rooms. In the first step, node attributes are decided by both geometric and semantic information of objects and edges determined based on facing formation. The second step is clustering each GSAG and finding an optimal common cluster that maximizes the weighted sum of matched edges and minimizes geometric dissimilarity. Through a collaboration space allocation, the registered cluster can be found among optimal common cluster pairs that can be interacted with. The generated shared virtual space, which reflects the common clusters, enables interactions based on multiple objects between dissimilar spaces.

Table 1: Sufficient conditions for edge creation vary depending on the affordance combination of the start node and end node. A facing angle is the minimum condition of the angle between the selected facing vectors of two nodes. $D_{adjacent}$ is a sufficient condition added in relation to (e) and (f). In this study, we set $\theta_{adjacent} = 45^\circ$, $\theta_{basic} = 135^\circ$, and $D_{adjacent} = 0.1m$.

	Edge Type	Min Facing angle	Max Distance
(a)	sittable-sittable	$\theta_{adjacent}$	-
(b)	sittable-placeable	θ_{basic}	-
(c)	displayable-sittable		
(d)	displayable-placeable	-	$D_{adjacent}$
(e)	placeable-placeable		
(f)	displayable-displayable	$\theta_{adjacent}$	$D_{adjacent}$

3.1 GSAG of a Single Room

3.1.1 Node decision

GSAG consists of nodes that correspond one-to-one with each object and edges, which are the relationships between them. GSAG uses a total of four affordances, a combination of three basic affordances (sittable, placeable, displayable) and leanable, which is dependent on sittable. Each affordance represents the action possibilities of sitting, placing, displaying, and leaning. Three basic affordances were used in research by Keshavarzi et al. [17] and Kim et al. [19, 22] to reflect user interactions with physical objects that are most likely to occur during remote collaboration, in terms of user position (sit) and action (place, display). Geometric spatial affordance based on the node's affordance and geometric information determines the facing vector that represents the interaction direction of the object. The facing vector represents the direction in which an object's action is possible. The plane where the action specified by affordance occurs is the action surface, and the vector perpendicular to it is the action vector, which indicates the facing direction of the action. The plane that limits the direction of action is called the constraint surface, and the vector perpendicular to it is the constraint vector. The action vector and the constraint vector are used to determine the facing vector considering affordances (Fig. 3). The specific details of the facing vector in each affordance type are as follows.

Sittable Sitting action takes place on the horizontal plane of the seat. The facing vector of a sittable object is four direction vectors perpendicular to each side of the action vector and the edge of the action surface. Multi-person seats are divided into nodes for each single-person seat. For example, in the case of a 3-person seat, it is divided into three nodes, and a facing vector is formed for each, following the same method as in the case of a single-person seat.

Leanable When a sittable object is also leanable, users can only face a certain direction when they sit on it. The plane that generates this constraint is called a constraint surface, and the constraint vector is a vector perpendicular to this plane. If a constraint vector exists, only the facing vector parallel to the constraint vector is valid. Multi-person seats that are leanable have two facing vectors per node, assuming that a virtual constraint vector occurs along the long side.

Placeable Placeable objects have a horizontal action surface, like a sittable. This applies to furniture that has a horizontal surface that allows the action of placing something, such as a desk or table. It has a facing vector perpendicular to the edge of the action surface and the action vector.

Displayable For displayable objects, the action surface is vertical, not horizontal. When content is displayed on an action surface, users can view displayable objects in a direction perpendicular to it. Therefore, both the action vector and the facing vector are perpendicular to the action surface and parallel to each other.

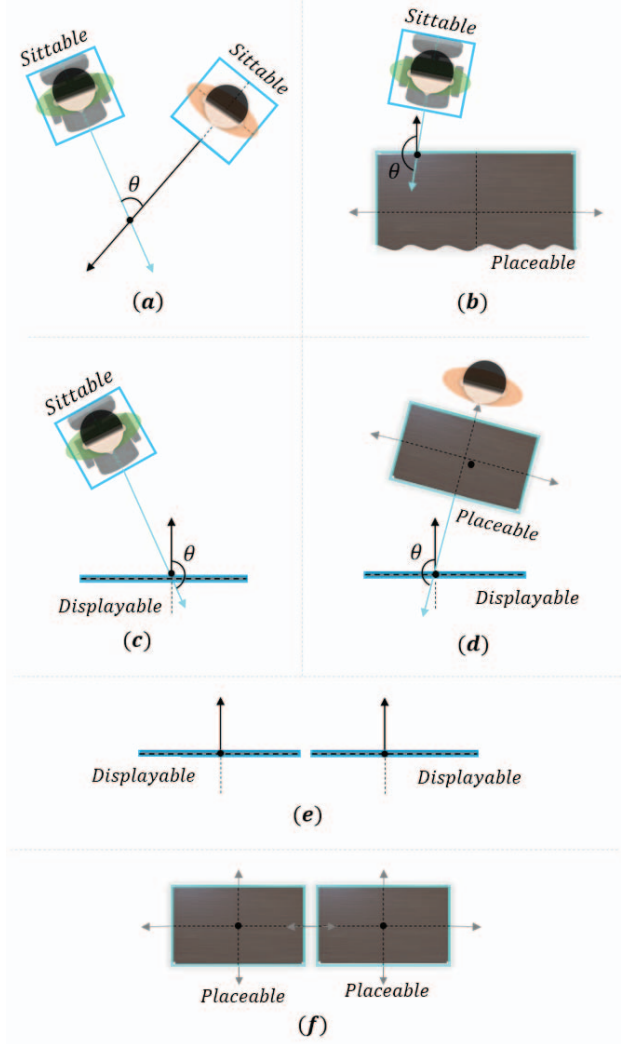


Figure 4: Example scenes of six different types of relations: (a) sittable-sittable, (b) sittable-placeable, (c) displayable-sittable, (d) displayable-placeable, (e) placeable-placeable, (f) displayable-displayable.

3.1.2 Edge generation

The edge of GSAG expresses the relationship between two objects, and its creation is determined based on the facing formation that reflects the locational characteristics for people to interact with each other [41]. In GSAG, an edge is created if there is a formation that allows furniture to interact with each other based on their location and direction, which represents the possibility of multi-object interaction. Among the facing vectors of the edge's start and end nodes, the facing vector perpendicular to the edge of the closest action surface is used to create the edge. An edge is created when the opposing vectors of each node point toward each other, and the facing criterion varies depending on the edge type (Table 1, Fig. 4). The minimum angle for facing is θ_{basic} and is set to 135° considering face-to-face formation. Edges between sittable nodes apply $\theta_{adjacent}$ (45°) to allow both L-shaped formation and face-to-face formation. For placeable objects to be connected and used together as a multi-object interaction, they must meet the maximum distance $D_{adjacent}$ ($0.1m$). Edges between displayables have connectivity by

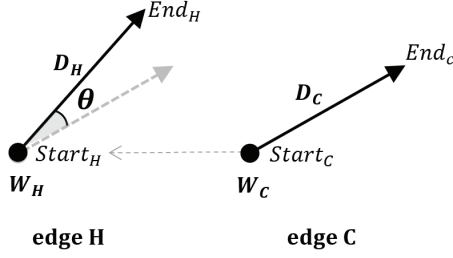


Figure 5: Calculation of geometric dissimilarity between matched edges in GSAG for $\psi_{geometric}(G_H, G_C)$.

satisfying both of the previous two conditions.

The weight for a graph G , $W(G)$, is the sum of the weight values of each edge, w_{edge} , calculated from the facing vector angle and Manhattan distance. The method is formulated as follows:

$$w_{edge}(i, j) = \omega_A \hat{A}(i, j) + \omega_D \hat{D}(i, j). \quad (1)$$

$$W(G) = \sum_{i,j=1}^{n(G)} w_{edge}(i, j). \quad (2)$$

When there are n nodes in a graph G , $\hat{A}(i, j)$, is the facing vector angle between the facing vector in start node i , and the facing vector in end node j is normalized by dividing by maximum value, 180. i and j less than the number of nodes in graph G are positive integers. $\hat{D}(i, j)$ is the Manhattan distance between i and j , which is normalized using the maximum and minimum distance values among all edges in graph G . Through this, importance is determined based on the relative difference between distances. The coefficients ω_A and ω_D are the weights in each term. The relative distance considered the detour of user movement using the Manhattan distance between two objects. When the displayable node was at the start or end node, an offset (1 m) was corrected to take long-distance interactivity into account.

3.2 OCR between asymmetric rooms

After creating the GSAG for each room, the optimal cluster pair can be found in the asymmetric space through OCR. The cluster found the optimal partition that promoted modularity optimization through the Louvain Community Detection Algorithm [4]. Two main goals decide the optimal cluster pair between GSAG. The first goal is to secure as many common relationships as possible. This leads to the possibility of various interactions in the shared virtual spaces. However, since more relationships are matched, more geometric dissimilarity between objects within the secured area inevitably occurs. Therefore, the second goal is to minimize the resulting geometric dissimilarity while accomplishing the first goal. The process of finding the optimal solution between two conflicting goals is the core of OCR and is expressed as the objective function below:

$$O(G_H, G_C) = \omega_r \psi_{relation}(G_H, G_C) - \omega_g \psi_{geometric}(G_H, G_C). \quad (3)$$

Given the host graph G_H and the client graph G_C , $\psi_{relation}$ is a term of how many relationships are matched in the two given graphs. At the same time, geometric dissimilarity calculated as the difference in direction and distance between matched edges must be minimized. Details of each term are as follows. Both ω_r and ω_g are positive integers less than 1.

Weight sum of matched clusters $\psi_{relation}(G_H, G_C)$ calculates how much the sum of matched edges accounts for the sum of the

weight of the entire graphs based on Jaccard similarity. In Equation 4, the weighted sum of matched clusters, g_H and g_C are divided by the total weight sum of given graphs, G_H and G_C . For g_H and g_C , all subgraphs of G_H and G_C are candidates.

$$\psi_{relation}(G_H, G_C) = \frac{W(g_H) + W(g_C)}{W(G_H) + W(G_C)}. \quad (4)$$

Geometric dissimilarity Geometric dissimilarity $\psi_{geometric}$ consists of angle dissimilarity and distance dissimilarity. In Equation 5, $\hat{A}(i, j)$ is the sum of values normalized by dividing the angle difference between matched edge pairs, (i, j) , in the matched cluster by 180 degrees. The angle between edges within each graph is calculated as the angle difference between vectors from the start node to the end node of each edge (Fig. 5).

$$\psi_{geometric}(G_H, G_C) = \sum_{i_h, j_h, i_c, j_c=1}^N (w_{edge}(i_h, j_h) - w_{edge}(i_c, j_c)). \quad (5)$$

N is number of nodes in g_H and it is also same as number of nodes in g_C . Both g_H and g_C is same as in Equation 5. Normalized distance dissimilarity, $\Delta \hat{D}$ is calculated based on the distance difference between matched edge pairs within the matched cluster. The theoretical minimum value of $\Delta \hat{D}$ is 0, and the maximum value is infinite. Considering this, by using the Sigmoid function, the closer $\Delta \hat{D}$ is to 0, the lower the positional dissimilarity between the two edges is, so the value gets closer to 1. Conversely, as the term value becomes larger, it means that they have high positional dissimilarity, so it approaches 0.

Based on optimal cluster pairs, a shared virtual space that secures relationships between objects can be formed. The existing shared space formation method adds all object areas with the same affordance when overlaid and measures the area with the largest final area [18, 21, 22]. In this case, all objects in the space have equal importance. The core of the proposed shared space generation through OCR aims to target registered optimal cluster pairs and maximize the overlapping area of the corresponding objects. When maximizing the area, the space is searched in all four directions. In addition, in the proposed method, a condition was added that the allocated object area must be at least 0.1 m^2 or at least 10% of the original area of the overlapping object for interactions to be possible.

Even if optimal cluster pairs are found and considered through OCR, the edges or nodes that were matched during the OCR process may not be assigned to the generated shared space due to geometric dissimilarity. Therefore, this study proposes Registered Edge-IoU (REI) as a novel spatial indicator to evaluate and analyze the performance of OCR using GSAG.

4 EVALUATION

We developed a novel approach to generate GSAG based on the geometric spatial affordance of objects and to perform OCR between asymmetric spaces using it. In evaluation, we show that the proposed OCR using GSAG is effective in considering relationships between objects and reflecting them in the shared virtual space. Since there were no quantitative indicators that measure how many relationships are secured in generated shared space, we suggest REI and use it with area size-related indicators that were previously used.

4.1 Experiment Design

The experiment compares a shared virtual space created by maximizing the semantic segmentation matching area with a shared virtual space created through the same process by targeting an object cluster registered by OCR. After creating a GSAG of a single space for a given space in the dataset, the optimal cluster pairs resulting from OCR for asymmetric space pairs are considered the target object

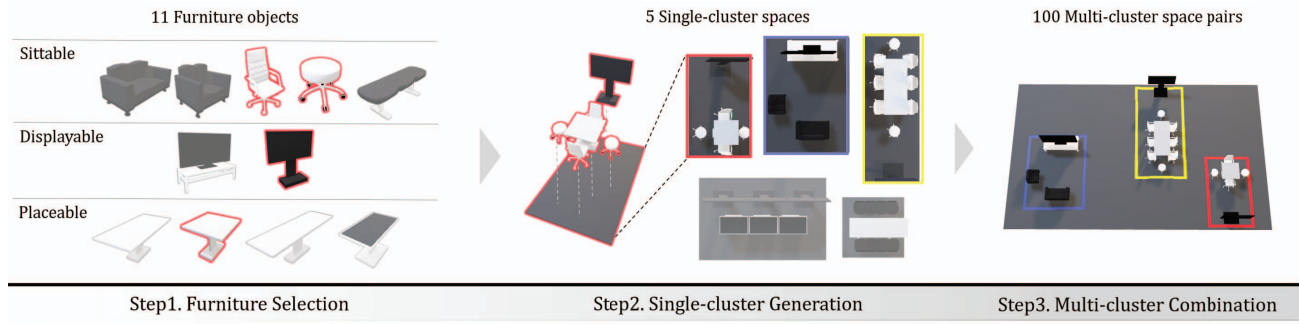


Figure 6: The figure shows the input space generation process for evaluating the proposed method. In the first step, objects of at least two different affordance types are manually selected from 11 furniture objects. In the second step, selected furniture objects are placed based on the digital twin of the real space to form a single cluster space. The five different single-cluster space configurations generated in step 2 were randomly combined into a multi-cluster space in step 3. As a result, 100 input space pairs were created and used for evaluation (10 host spaces $\times$ 10 client spaces).

area. In this experiment, OCR was performed on edges with weights greater than the overall average of normalized edge weight values for a single space GSAG. Afterward, rather than maximizing the area for the entire area where semantic segmentation matches, the area where semantic segmentation matches only for the target object area is maximized. In the case of other furniture objects that do not become the target object area, they are considered objects with unknown affordances and a low possibility of interaction in shared space generation. The final shared virtual space formation method follows an existing algorithm to determine the location and direction of overlap between spaces to maximize the area to be matched. Additionally, because the goal of this study is object interaction within the shared space, there is an additional condition for object interaction to be considered possible when the semantically matched space is at least the minimum size (0.1 m^2) or when more than 10% of the original object area is utilized.

The most essential and widely used quantitative indicator for measuring existing shared spaces was area. Many existing studies have evaluated how much area is registered in a given pair of spaces. However, there was no quantitative indicator to evaluate how well the relationships between objects for interaction in the created space were matched and registered. Therefore, this study proposes a quantitative indicator to measure how many or how major object relationships are included among the object relationships in each space through REI, and uses it to measure the results of the two methods. Below is the specific calculation method of REI.

Registered Edge-IoU $REI(G_H, G_C)$ evaluates how much of the relationship between objects is allocated in the final created space. The basic concept is similar to $\Psi_{relation}$ (Equation 4). However, while $\Psi_{relation}$ is the degree of relation preservation as a term of the objective function, REI indicates, actually, registered relation preservation between asymmetric spaces. REI is also based on Jaccard similarity, which is the weight sum of each registered cluster $W(g'_H)$ and $W(g'_C)$ is divided by the weight sum of both graphs, G_H and G_C (Equation 6).

$$REI(G_H, G_C) = \frac{W(g'_H) + W(g'_C)}{W(G_H) + W(G_C)}, \quad (6)$$

$$(g'_H \subset G_H, g'_C \subset G_C)$$

Registered Area-IoU $RAI(A_H, A_C)_{Type}$ calculates the size of the common area compared with the total area between two spaces while A_H indicates the area size of the host space. A_C indicates the area size of the client space and $Type$ indicates the type of area. There are four types of areas: Movable, Furniture, Cluster, and Total. The movable area is the floor area without any furniture. Furniture area is the sum of areas occupied by any furniture. The cluster area is the

sum of areas occupied by registered furniture. Finally, Total area is the sum of movable areas and furniture areas. RAI is also based on Jaccard similarity, and the common area size is divided by the union area size of two spaces (Equation 7).

$$RAI(A_H, A_C)_{Type} = \frac{(A_H \cap A_C)_{Type}}{A_H \cup A_C}, \quad (7)$$

$$(Type = \{Movable, Furniture, Cluster, Total\})$$

4.2 Input Space

For evaluation, we created multi-cluster spaces based on real indoor space digital twins and used them as input spatial data. This is because a large number of multi-cluster spaces are needed to test how well our method can find similar clusters in various spatial configurations. Input space generation is largely divided into three steps (Fig. 6). In the first step, among 11 furniture objects with one of four types of affordances (three basic affordances and leanable), three or more objects were selected to include at least two affordance types for spatial diversity in each input space. In the second step, selected furniture objects were placed based on the digital twin of the real indoor space to form a single-cluster space with five different spatial configurations. In the final step, a multi-cluster space containing two or more single clusters was created and used through random combination and arrangement of the generated single cluster space.

To increase the realism of the generated spaces, we developed a multi-cluster space generator with multiple rules applied in the Unity 2021.3.21f1 environment. The GSAG generator was developed to generate spatial scene graphs by placing affordance object assets based on real spatial data. Several constraints were set to ensure that the generated space reflected the real environment. For example, a maximum of three clusters can be set up in one room. In addition, considering the spacing between each object during placement, the floor's width was limited to less than 12 m. Relocation occurs when conflicts occur between clusters. An example of the five single-cluster spaces used and the multi-cluster spaces created can be found in Fig. 6. The created 20 multi-cluster spaces were divided into host groups and client groups. Ultimately 100 multi-cluster space pairs were created (10 host spaces $\times$ 10 client spaces).

4.3 Results

The results are illustrated in Fig. 7. Five indicators (REI, Movable RAI, Furniture RAI, Cluster RAI, and Total RAI) were compared between shared virtual spaces with and without OCR. They consist of four different RAI indicators and one REI indicator. The

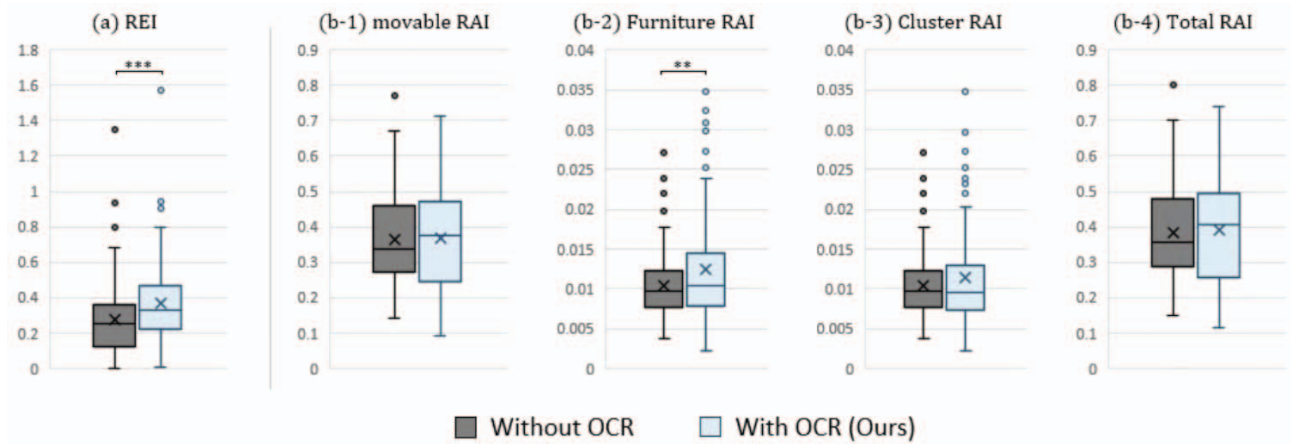


Figure 7: Five indicators (a) REI, (b-1) Movable RAI, (b-2) Furniture RAI, (b-3) Cluster RAI, (b-4) Total RAI were compared between the shared virtual space where OCR was used and the shared virtual space where OCR was not used. Black is a shared virtual space that only maximizes area without using OCR, and blue is a shared virtual space condition that seeks to maximize area after using OCR.

Shapiro-Wilk test was used to examine the normality of data distribution. Since all the data did not follow normality (all $p < .05$), the Wilcoxon Signed-Rank test ($\alpha = 0.05$) was utilized as a non-parametric measure to compare the differences in the results of the two methods on the common dataset.

We found significant differences in both REI and Furniture RAI (REI: $W = 886$, $Z = -4.24$, $p < .001$; Furniture RAI: $W = 1528$, $Z = -3.18$, $p < .01$). Regarding REI in Fig. 7(a), the condition with OCR was about 0.369, which was 34.19% higher than 0.275 in the condition without OCR. Through this, it was shown that the proposed OCR using GSAG is effective in preserving object relationships within the shared space formed. Among the four different RAI indicators in Fig. 7(b-2), only the RAI of furniture objects (Furniture RAI) was 0.012 with OCR and 0.010 without OCR, 19.34% higher.

However, for the other three RAI indicators, there were no statistically significant differences (Movable RAI: $W = 2360$, $Z = -0.40$, $p = .688$; Cluster RAI: $W = 2321$, $Z = -0.37$, $p = .711$; Total RAI: $W = 2242$, $Z = -0.81$, $p = .416$). Regarding the total shared area (Total RAI) in Fig. 7(b-4), there was no significant difference between the cases where OCR was used (0.391) and the cases where OCR was not used (0.384). Similarly, the shared movable area (Movable RAI) in Fig. 7(b-1) that affects the walking experience also showed no statistically significant difference; the condition using OCR was 0.368, and the condition not using OCR was about 0.365. Finally, regard to the shared area of furniture objects excluding obstacles (Cluster RAI) in Fig. 7(b-3) condition with OCR (0.011) and without OCR (0.010) has no significant difference.

Additionally, to verify the applicability of GSAG to real-life settings, we applied GSAG to the multi-cluster space of the Matterport 3D dataset as shown in Fig. 8. To ensure diversity across spaces in the Matterport 3D dataset, spaces with a variety of furniture objects and configurations were selected. Based on the selected spatial information, a scene graph was formed by arranging the furniture in consideration of the position and size measures from the previously created scene graph generator.

5 DISCUSSION

The proposed method aims to establish a shared virtual space that enables diverse and complex interactions with affordance-based object clusters. We devised the GSAG to represent relationships among objects within a single user space. This representation is then leveraged in our object cluster generation algorithm to identify optimal clusters of related objects and register them across different spaces.

To measure the performance of our GSAG-based OCR, we proposed a novel quantitative metric called the Relationship Expression Index (REI). We evaluated the effectiveness of this approach on two datasets, comparing it with the area indicator used in prior research.

In the study, four different spatial indicators and REI were used for evaluation. The shared virtual space formed through GSAG-based OCR had an REI that was approximately 34.19% higher than the space where it was not applied, and the RAI of the furniture object area was also 19.34% higher. On the other hand, there was no statistically significant difference in the RAI of other movable areas, cluster areas, and the total area combined. This indicates that even when the relationship between objects within a space is considered by maximizing the weighted sum of matched edges, the size of the total area generated is not affected. We infer that the reason why there was a significant increase for the entire furniture area when there was no significant difference for the object area within the targeted cluster is because our method focuses on only the target group area with high similarity rather than considering all the various furniture objects distributed in the room. Through these results, we demonstrated that our proposed method is capable of preserving and reflecting object relationships in the generation of shared virtual spaces without compromising the commonly sought-after goal of maximizing their area.

The multi-cluster space used in the evaluation consisted of a random combination and arrangement of five single-cluster spaces. Because the current method finds the optimal cluster between two spaces, the single cluster with the most edges matched by using this method is often determined as the final registered cluster. For this, we believe that our method can be further improved to identify adaptive optimal cluster pairs based on specific spatial conditions or collaboration scenarios. Additionally, we foresee the development of more efficient methods that can align highly dissimilar spaces by utilizing a list of two or more optimal cluster pairs that build upon our approach. It would also be beneficial to investigate how the matched space evolves concerning the number of clusters in the results and the arrangement or relationship between these clusters.

6 CONCLUSION AND FUTURE WORK

We proposed OCR using GSAG to generate shared virtual spaces that can support interaction with multiple objects for users collaborating remotely in dissimilar real spaces. The relationships between object clusters of dissimilar spaces can be expressed through GSAG. Based on the graph, optimal object cluster pairs can be found by comparing

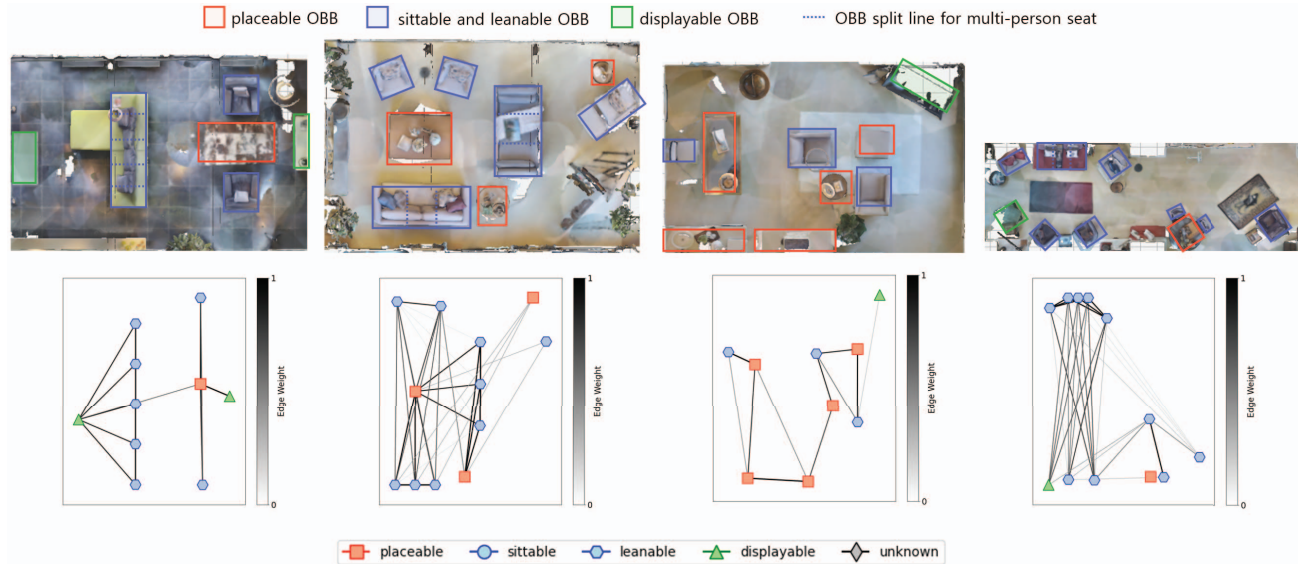


Figure 8: Four Matterport3D multi-cluster spaces (top) and the results of generating GSAG from them (bottom). Each object bounding box is marked on the top view of spaces.

the geometric dissimilarity and affordance similarity between each space through OCR. We compared the GSAG-based OCR method with one that does not for 100 pairs of multi-cluster spaces generated from random combinations of five different single clusters. The results demonstrated a 34.19% higher rate of relationship preservation for the OCR method based on GSAG, while there was no significant change in total area or movable area. We showed that the proposed method can create a shared virtual space for multi-object interaction without conflicting with the goal of existing methods of maximizing the total area. Additionally, we confirmed the real-space applicability of GSAG generation using the Matterport3D dataset

Our method enhances collaboration activities within a shared virtual space created from diverse real spaces. Nevertheless, acknowledging and addressing the limitations of our approach in future research will be crucial to cater to a broader range of collaboration scenarios and complex real-world spaces. First, we intend to increase the types of affordances integrated into the GSAG-based OCR, which will allow a greater variety of interactions within the shared virtual space. Second, we will verify the validity of our method in spaces that are highly dissimilar by further developing the GSAG to consider the dynamic movement of objects in real space or the addition of virtual objects. Third, we require a user evaluation to assess the impact of the shared virtual space generated by our GSAG-based OCR method on user experience during remote collaboration.

In the long term, our goal is to design an adaptive multi-cluster matching method utilizing extended spatial indexes and validate it across a broader array of multi-cluster space types. Our vision is that representing space as an intricate structure with interconnected objects and identifying common object clusters will serve as the foundation for user-oriented generations of shared virtual spaces. These spaces should support a wide range of complex and diverse scenarios for remote collaboration in dissimilar spatial environments. Our method can potentially raise the level of realism and immersion by enabling users to interact with actual tables and chairs in their real, individual environments to engage in active movement within the same virtual, shared setting.

ACKNOWLEDGMENTS

This research was supported by the National Research Council of Science and Technology(NST) funded by the Ministry of Science and ICT(MSIT), Republic of Korea(Grant No. CRC 21011) and the MSIT(Ministry of Science and ICT), Korea, under the Graduate School of Metaverse Convergence support program(IITP-2024-RS2022-00156435) supervised by the IITP(Institute for Information & Communications Technology Planning & Evaluation).

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